Dear Ariel,

I really enjoyed reading your post. It effectively connects real-world industrial applications with broader Industry 4.0 concepts such as smart factories, predictive maintenance, and quality optimisation through the experience of a leading automotive manufacturer.

I particularly agree with your emphasis on predictive maintenance, which has become a central pillar of Industry 4.0 due to its ability to reduce unexpected downtime, improve equipment availability, and support data-driven decision-making within manufacturing systems (Zonta *et al.*, 2020).

Your discussion of enhanced product quality through Quality 4.0 is also highly relevant in the automotive sector, where AI-driven quality control contributes to vehicle reliability, organisational reputation, customer loyalty, and reduced warranty costs (Zhou *et al.*, 2019).

Your example of the Jaguar Land Rover cybersecurity incident clearly highlights the broad impact such events can have on Industry 4.0 organisations. Although the affected stage of the manufacturing process was not specified, the disruption may have originated from a supply chain vulnerability, given the critical role of interconnected suppliers and third-party systems in Industry 4.0 environments (Zarreh *et al.*, 2019).

To reduce the likelihood of similar incidents, manufacturing leaders recommend evidence-based cybersecurity approaches, including continuous risk assessments and adherence to industry standards (Okomanyi, Sherwood and Shittu, 2024). Among these, the National Institute of Standards and Technology Cybersecurity Framework (NIST CSF) is recognised as one of the leading frameworks for cybersecurity assurance in the automotive industry (Burkacky et al, 2020).

In addition, specific frameworks have been proposed to address supply chain security concerns within Industry 4.0 environments. Sowan, Zighan and Altarawneh (2025) proposed a multi-layered framework that integrates technologies such as blockchain, AI, IoT, and big data analytics to strengthen security through decentralised data management, real-time monitoring, and automated threat detection. Their framework also combines technological solutions with organisational strategies, including risk assessment, employee training, and collaborative risk management.

To conclude, several approaches can be adopted to strengthen cybersecurity assurance. Most industry leaders recommend well-established frameworks such as the NIST Cybersecurity Framework (NIST CSF), while emerging supply chain security challenges may require newer frameworks grounded in Industry 4.0 technologies.

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